Main Content

1. Trigonometric Form of Fourier Series
2. Exponential Form of Fourier Series
3. Relationship Between Function Symmetry and Fourier Series Coefficients
4. Parseval's Theorem
5. Finite Fourier Series and Minimum Mean Square Error
6. Spectral Structure of Periodic Rectangular Pulse Trains
7. Gibbs Phenomenon
8. Bandwidth

1. Trigonometric Form of Fourier Series

Periodic signal $f(t)$ Fundamental period T_0 Fundamental angular frequency $\omega_0 = \frac{2\pi}{T_0}$

The signal $f(t)$ can be decomposed into a combination of harmonically related trigonometric functions

$$f(t) = a_0 + \sum_{k=1}^{\infty} \left[a_k \cos(k\omega_0 t) + b_k \sin(k\omega_0 t) \right]$$

$$\{\cos(k\omega_0 t), \sin(k\omega_0 t)\}, \quad k=0, 1, 2, \cdots$$

This set forms a complete orthogonal function set over one period T_0

$f(t)$ must satisfy the Dirichlet conditions (ensuring the series converges to the signal)

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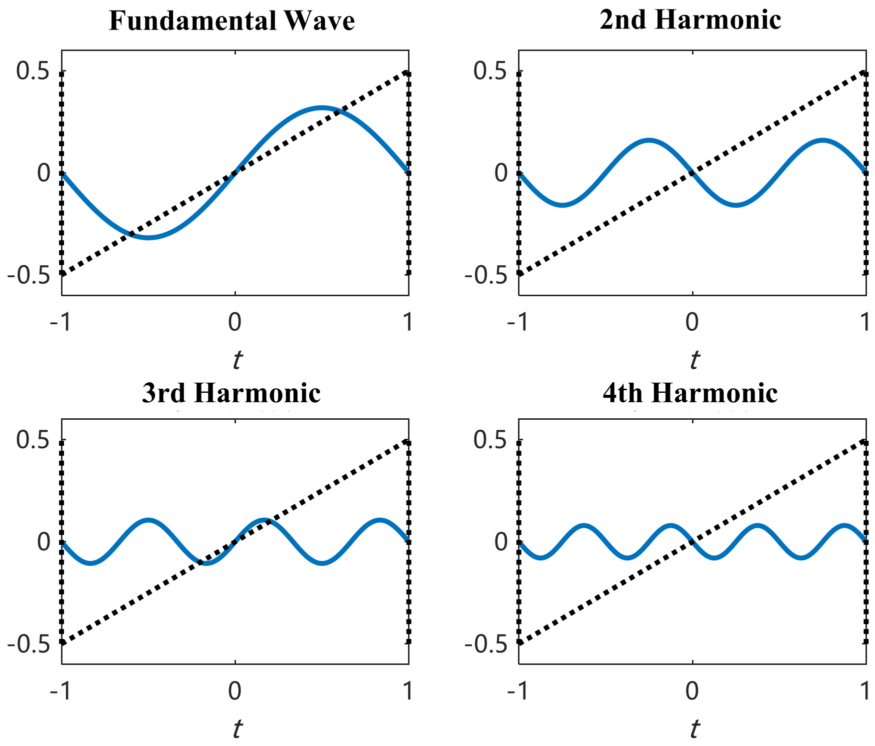
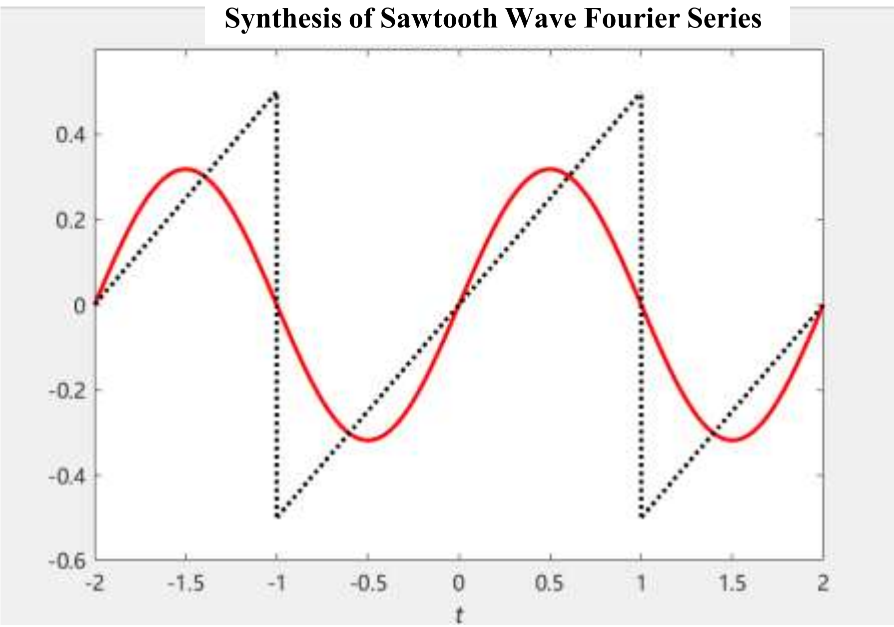
(1) Harmonically Related





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(2) Complete Orthogonal Function Set

$\{\cos(k\omega_0 t), \sin(k\omega_0 t)\}$, k=0, 1, 2,$\cdots$ is orthogonal function set

$$\int_0^{T_0} \sin(n\omega_0 t)\sin(m\omega_0 t)\,\mathrm{d}t = \begin{cases} 0 & n \neq m \\ \frac{1}{2}T_0 & n = m \end{cases}$$

$$\int_0^{T_0} \cos(n\omega_0 t)\cos(m\omega_0 t)\,\mathrm{d}t = \begin{cases} 0 & n \neq m \\ \frac{1}{2}T_0 & n = m \end{cases}$$

$$\int_0^{T_0} \sin(n\omega_0 t)\cos(m\omega_0 t)\,\mathrm{d}t = 0$$

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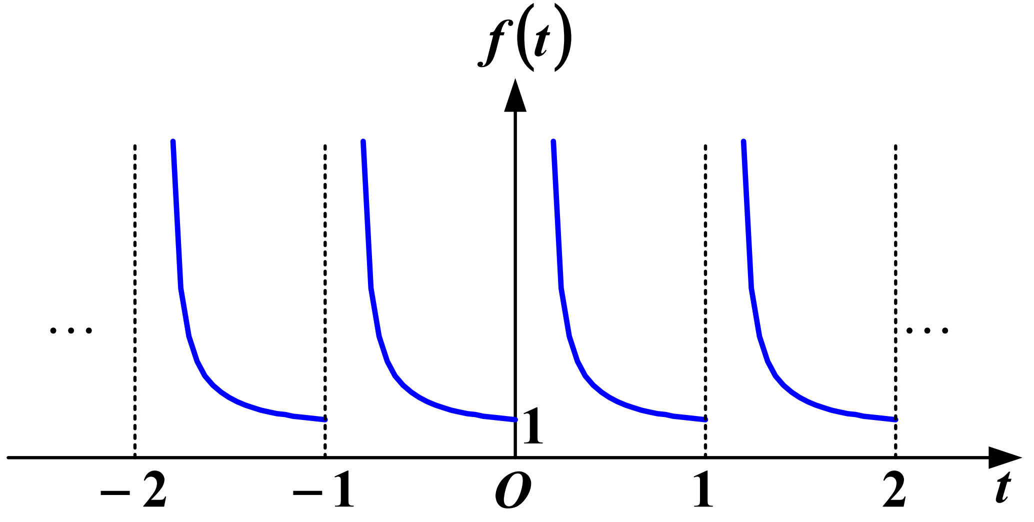
(3) Dirichlet conditions

Condition 1: Absolute Integrability Over One Period

$$\int_{t_0}^{t_0+T_0} \left| f(t) \right| \mathrm{d}t < \infty \qquad (T_0, \text{ period})$$

For a signal with a period of 1, if the signal within the interval ($0 < t \leq 1$) is $f(t)=\frac{1}{t}$

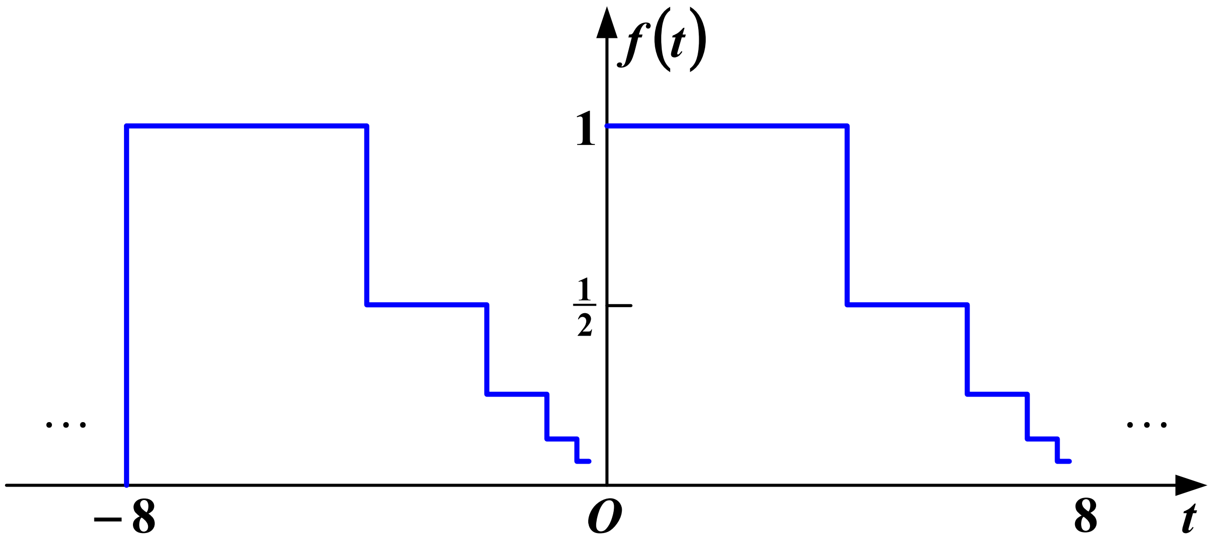


Fails to satisfy the absolute integrability condition

(3) Dirichlet conditions

Condition 2: Within a single period, if discontinuity points exist, their number must be finite

Example violating Condition 2: This signal has a period of 8, where each subsequent step's height and width are half of the previous step. Although its total area within one period does not exceed 8, the number of discontinuity points is infinite

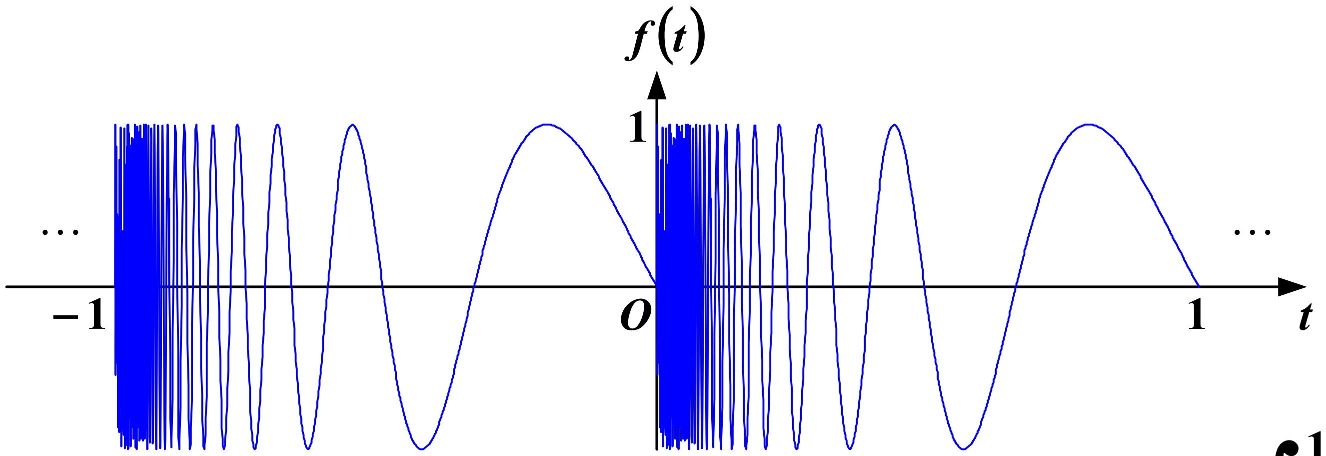


(3) Dirichlet conditions

Condition 3: Within one period, the number of local maxima and minima must be finite

A function violating Condition 3:

$$f(t)=\sin\left(\frac{2\pi}{t}\right),(0<t\leq 1)$$



For comparison, this periodic function (with period 1) satisfies $\int_0^1 |f(t)| \mathrm{d}t < 1$

(4) How to determine the expansion coefficients?

$$f(t) = a_0 + \sum_{k=1}^{\infty}\left[a_k \cos\left(k\omega_0 t\right) + b_k \sin\left(k\omega_0 t\right)\right]$$

Review: Real Orthogonal Function Decomposition

$$f(t) = c_1 g_1(t) + c_2 g_2(t) + \ldots + c_r g_r(t) + \cdots$$

Coefficient

$$c_r = \frac{\int_{t_1}^{t_2} f(t) g_r(t) \,\mathrm{d}t}{\int_{t_1}^{t_2} g_r^2(t) \,\mathrm{d}t}$$

(4) How to determine the expansion coefficients?

$$f(t)=a_0+\sum_{k=1}^{\infty}\left[a_k\cos\left(k\omega_0 t\right)+b_k\sin\left(k\omega_0 t\right)\right]$$

$$a_0=\frac{1}{T_0}\int_{t_0}^{t_0+T_0} f(t)\,\mathrm{d}t$$

The average value of a signal

$$a_k=\frac{\int_{t_0}^{t_0+T_0} f(t)\cos\left(k\omega_0 t\right)\mathrm{d}t}{\int_{t_0}^{t_0+T_0}\cos^2\left(k\omega_0 t\right)\mathrm{d}t}=\frac{\int_{t_0}^{t_0+T_0} f(t)\cos\left(k\omega_0 t\right)\mathrm{d}t}{\int_{t_0}^{t_0+T_0}\frac{1+\cos\left(2k\omega_0 t\right)}{2}\mathrm{d}t}$$

The integral of the double frequency term is 0

$$=\frac{2}{T_0}\int_{t_0}^{t_0+T_0} f(t)\cos\left(k\omega_0 t\right)\mathrm{d}t$$

Similarly $$b_k=\frac{2}{T_0}\int_{t_0}^{t_0+T_0} f(t)\sin\left(k\omega_0 t\right)\mathrm{d}t$$

(5) Harmonic Form (Frequency-Term Merging)

$$f(t) = a_0 + \sum_{k=1}^{\infty}\left[a_k \cos\left(k\omega_0 t\right) + b_k \sin\left(k\omega_0 t\right)\right]$$

Merged Harmonic Form (Cosine-Phase Representation)

$$f(t) = c_0 + \sum_{k=1}^{\infty} c_k \cos\left(k\omega_0 t + \varphi_k\right)$$

$k = 0,\quad c_0 = a_0$ DC Component, Average Value

$k = 1$ Fundamental wave, also called the 1st harmonic in some literature

k k-th harmonic

(5) Harmonic Form (Frequency-Term Merging)

$$a_k \cos(k\omega_0 t) + b_k \sin(k\omega_0 t)$$
$$= c_k \cos(k\omega_0 t + \varphi_k)$$
$$= c_k \cos(k\omega_0 t)\cos(\varphi_k) - c_k \sin(k\omega_0 t)\sin(\varphi_k)$$

$$\boxed{\cos(\alpha+\beta) = \cos(\alpha)\cos(\beta) - \sin(\alpha)\sin(\beta)}$$

Compare Coefficients, get $a_k = c_k \cos(\varphi_k)$ $\quad b_k = -c_k \sin(\varphi_k)$

Solve for c_{k} and φ_k

$$c_k^2 = a_k^2 + b_k^2$$

$$\tan(\varphi_k) = -\frac{b_k}{a_k}$$

$$c_k = \sqrt{a_k^2 + b_k^2} \quad (c_k \geq 0)$$

$$\varphi_k = \arctan\left(-\frac{b_k}{a_k}\right)?$$

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Example 1

Given $f_1(t) = \sin\omega_0 t,\ \ f_2(t) = \sqrt{3}\cos\omega_0 t,$ combining the two same frequency terms to get $f(t) = f_1(t) + f_2(t)$

$$a_1 = \sqrt{3}, b_1 = 1,\ \ \text{Thus } c_1 (\geq 0)$$

$$c_1 = \sqrt{a_1^2 + b_1^2} = 2$$

$$\varphi_1 = -\arctan\frac{b_1}{a_1} = -\arctan\frac{1}{\sqrt{3}} = -\frac{\pi}{6} \quad (\text{rad/s})$$

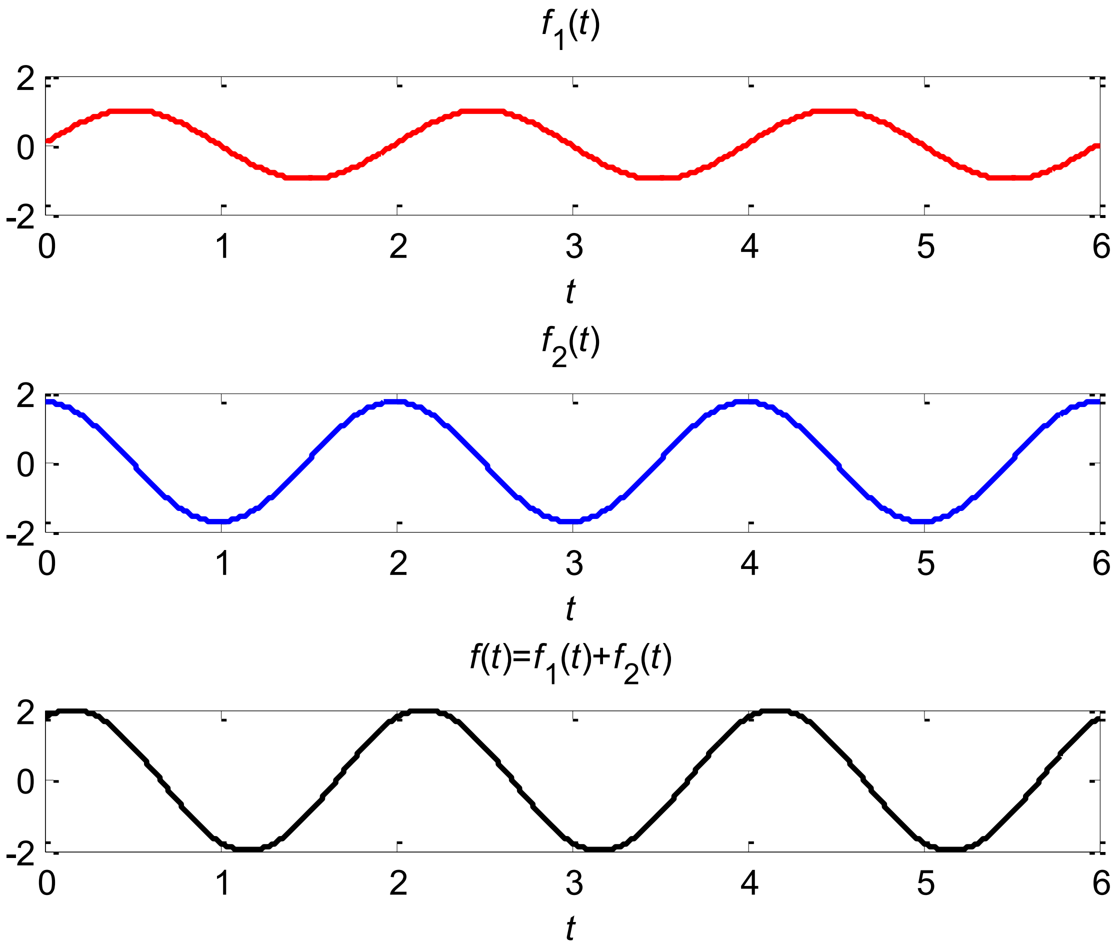
$$f(t) = 2\cos\left(\omega_0 t - \frac{\pi}{6}\right)$$

Example 1

$$f_1(t) = \sin \omega_0 t$$

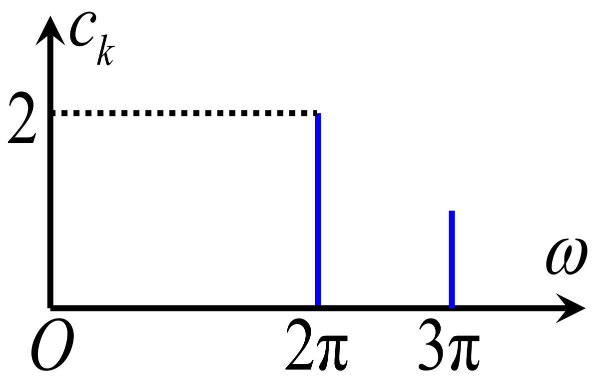
$$f_2(t) = \sqrt{3} \cos \omega_0 t$$

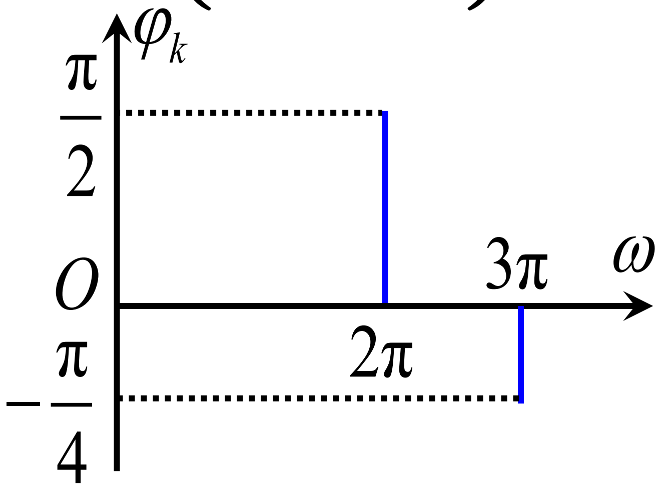
$$f(t) = f_1(t) + f_2(t)$$



(6) Spectrum Diagram

$$f(t) = 2\cos\left(2\pi t + \frac{\pi}{2}\right) + \cos\left(3\pi t - \frac{\pi}{4}\right)$$





$c_k \sim \omega$ **curve is called the amplitude spectrum diagram**

$\varphi_k \sim \omega$ **curve is called the phase spectrum diagram**

2. Exponential Form of Fourier Series

(1) From Trigonometric Functions to Exponential Functions

$$f(t) = c_0 + \sum_{k=1}^{\infty} c_k \cos\left(k\omega_0 t + \varphi_k\right)$$

$$c_k \cos\left(k\omega_0 t + \varphi_k\right)$$

Euler's formula $\quad \dfrac{1}{2} c_k \left[\mathrm{e}^{\mathrm{j}\left(k\omega_0 t + \varphi_k\right)} + \mathrm{e}^{-\mathrm{j}\left(k\omega_0 t + \varphi_k\right)} \right]$

$$= \frac{1}{2} c_k \mathrm{e}^{\mathrm{j}\varphi_k} \mathrm{e}^{\mathrm{j}k\omega_0 t} + \frac{1}{2} c_k \mathrm{e}^{-\mathrm{j}\varphi_k} \mathrm{e}^{-\mathrm{j}k\omega_0 t}$$

$$= F_k \mathrm{e}^{\mathrm{j}k\omega_0 t} + F_{-k} \mathrm{e}^{-\mathrm{j}k\omega_0 t}$$

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(2) Exponential Form of Fourier Series Expression

$\left\{ e^{jk\omega_0 t} \right\}, k = 0, \pm1, \pm2 \cdots$ is orthogonal function set

Series form

$$f(t) = \sum_{k=-\infty}^{\infty} F_k \, e^{jk\omega_0 t}$$

Fourier series coefficients

It can also be written as $F\left(k\omega_0\right)$

$$F_k = \frac{\int_{t_0}^{t_0+T_1} f(t)\, e^{-jk\omega_0 t}\, dt}{\int_{t_0}^{t_0+T_1} e^{jk\omega_0 t} e^{-jk\omega_0 t}\, dt}$$

Projection coefficient

$$= \frac{1}{T_1} \int_{t_0}^{t_0+T_1} f(t) e^{-jk\omega_0 t} dt$$

(3) Amplitude Spectrum and Phase Spectrum

$$f(t)=\sum_{k=-\infty}^{\infty} F_k \, e^{jk\omega_0 t}$$

F_k is a complex coefficient and can be expressed in polar form as $F_k = |F_k| e^{j\varphi_k}$

$$f(t)=\sum_{k=-\infty}^{\infty} |F_k| e^{j\varphi_k} \, e^{jk\omega_0 t} = \sum_{k=-\infty}^{\infty} |F_k| e^{j(k\omega_0 t+\varphi_k)}$$

$|F_k| \sim \omega$ **Amplitude spectrum**

$\varphi_k \sim \omega$ **Phase spectrum**

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(4) Relationship Between the Two Forms of Fourier Series

Assuming $f(t)$ is a real-valued function and $k>0$

$$F_k = \frac{1}{T_1}\int_{T_1} f(t)\mathrm{e}^{-\mathrm{j}k\omega_0 t}\,\mathrm{d}t$$

$$= \frac{1}{T_1}\int_{T_1} f(t)\cos\left(k\omega_0 t\right)\mathrm{d}t - \mathrm{j}\frac{1}{T_1}\int_{T_1} f(t)\sin\left(k\omega_0 t\right)\mathrm{d}t$$

$$= \frac{1}{2}\left(a_k - \mathrm{j}b_k\right)$$

$$F_{-k} = \frac{1}{T_1}\int_{T_1} f(t)\cos\left(k\omega_0 t\right)\mathrm{d}t + \mathrm{j}\frac{1}{T_1}\int_{T_1} f(t)\sin\left(k\omega_0 t\right)\mathrm{d}t$$

$$= \frac{1}{2}\left(a_k + \mathrm{j}b_k\right)$$

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(4) Relationship Between the Two Forms of Fourier Series

$$\begin{cases} F_k = \frac{1}{2}(a_k - \mathrm{j}b_k) \\ F_{-k} = \frac{1}{2}(a_k + \mathrm{j}b_k) \end{cases} \qquad \begin{cases} F_k = |F_k|\mathrm{e}^{\mathrm{j}\varphi_k} \\ F_{-k} = |F_{-k}|\mathrm{e}^{\mathrm{j}\varphi_{-k}} \end{cases}$$

Amplitude-Frequency Characteristic

$$|F_k| = |F_{-k}| = \frac{1}{2}\sqrt{a_k^2 + b_k^2} = \frac{1}{2}c_k$$

Phase-Frequency Characteristic

$$\varphi_k = \arctan\left(-\frac{b_k}{a_k}\right)$$

$$\varphi_k = -\varphi_{-k}$$

$$\varphi_{-k} = \arctan\left(\frac{b_k}{a_k}\right)$$

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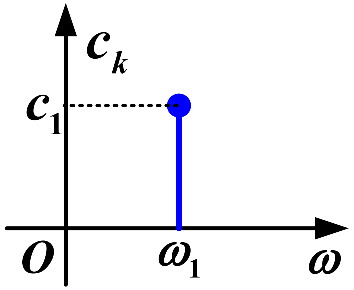
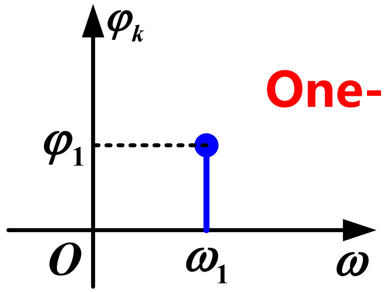
Example 2

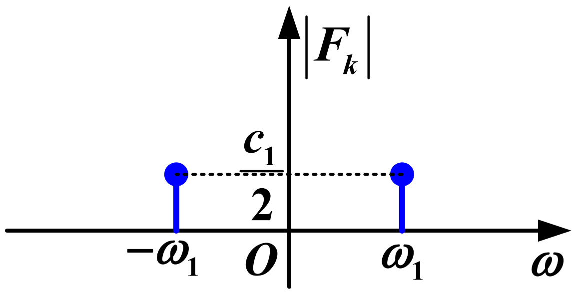
$$f(t) = c_1\cos(\omega_1 t + \varphi_1) = \frac{1}{2}c_1\left[e^{-j(\omega_1 t+\varphi_1)} + e^{j(\omega_1 t+\varphi_1)}\right]$$

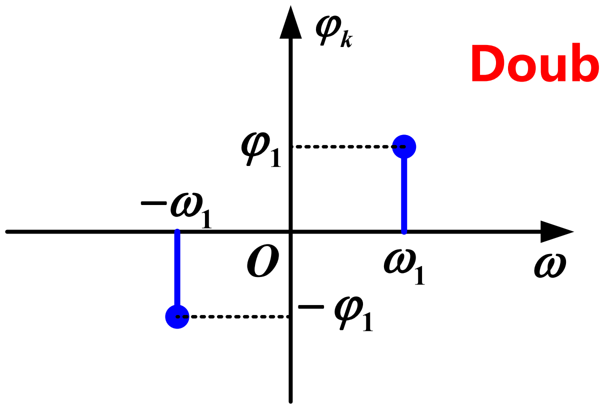




One-Sided Spectrum

$$f(t) = \frac{1}{2}c_1 e^{-j\varphi_1}e^{-j\omega_1 t} + \frac{1}{2}c_1 e^{j\varphi_1}e^{j\omega_1 t}$$





Double-Sided Spectrum

Example 3

$$f(t) = 1 + \sin\omega_0 t + \sqrt{3}\cos\omega_0 t - \cos\left(2\omega_0 t + \frac{5\pi}{4}\right)$$

Plot the Amplitude spectrum and phase spectrum

Let $f_1(t) = \sin\omega_0 t + \sqrt{3}\cos\omega_0 t$

Merge terms with the same frequency and convert to cosine form

Utilizing the result from Example 1

$$f_1(t) = 2\cos\left(\omega_0 t - \frac{\pi}{6}\right)$$

$$-\cos\left(2\omega_0 t + \frac{5\pi}{4}\right)$$

$$= \cos\left(2\omega_0 t + \frac{5\pi}{4} - \pi\right)$$

$$= \cos\left(2\omega_0 t + \frac{\pi}{4}\right)$$

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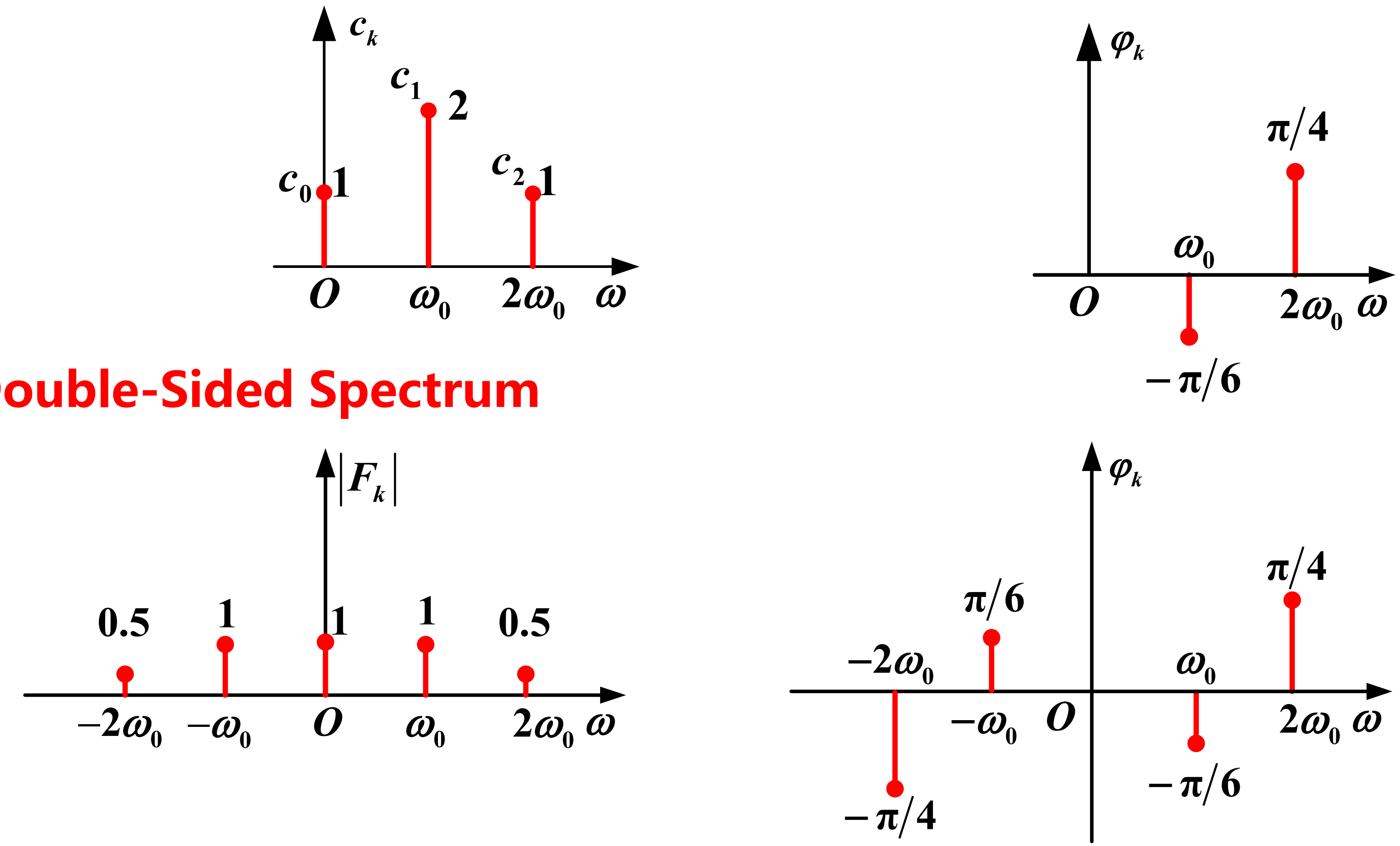
Example 3

$$f(t) = 1 + 2\cos\left(\omega_0 t - \frac{\pi}{6}\right) + \cos\left(2\omega_0 t + \frac{\pi}{4}\right)$$

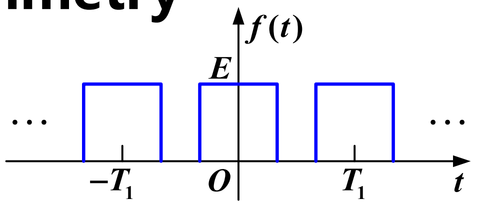
One-Sided Spectrum

Double-Sided Spectrum

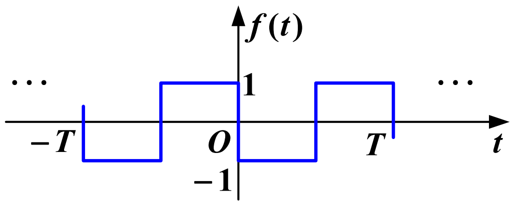


3. Relationship Between Function Symmetry and Fourier Series Coefficients

Whole-cycle symmetry

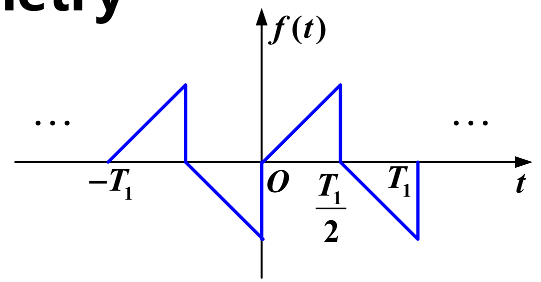


$$f(t) = f(-t)$$

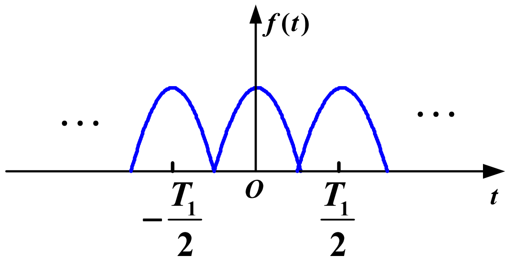


$$f(t) = -f(-t)$$

Half-cycle symmetry



$$f(t) = -f\left(t \pm \frac{T_1}{2}\right)$$

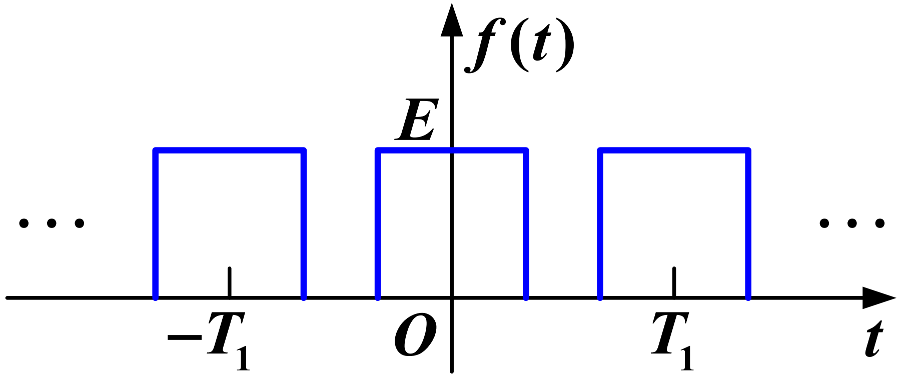


$$f(t) = f\left(t \pm \frac{T_1}{2}\right)$$

(1) Even Function

A signal waveform is symmetric with respect to the y-axis

$$f(t) = f(-t)$$

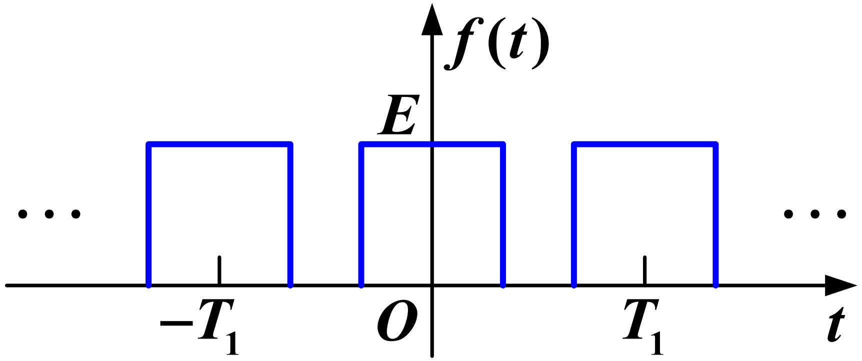


The sine function is an odd function, so even functions do not have odd function components

$$b_k = 0$$

$$a_k = \frac{4}{T_1}\int_0^{\frac{T_1}{2}} f(t)\cos(k\omega_0 t)\,\mathrm{d}t \neq 0$$

(1) Even Function



$$b_k = 0 \qquad a_k \neq 0$$

$$F_k = \frac{1}{2}\left(a_k - \mathrm{j}b_k\right) = \frac{1}{2}a_k$$

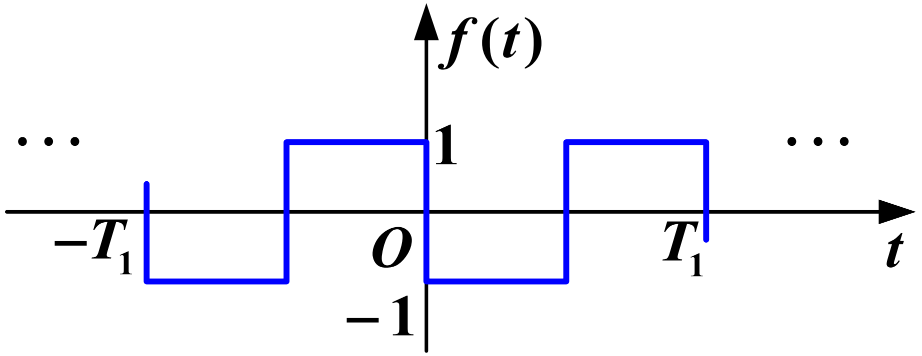
F_k is a **real** function

Time-domain even function ➔ Frequency-domain real function

(2) Odd Function

The waveform is antisymmetric about the vertical axis

$$f(t) = -f(-t)$$



$$a_0 = \frac{1}{T_1}\int_{-\frac{T_1}{2}}^{\frac{T_1}{2}} f(t)\,\mathrm{d}t = 0$$

$$a_k = \frac{2}{T_1}\int_{-\frac{T_1}{2}}^{\frac{T_1}{2}} f(t)\cos(k\omega_0 t)\,\mathrm{d}t = 0$$ No cosine term

$$b_k = \frac{4}{T_1}\int_{0}^{\frac{T_1}{2}} f(t)\sin(k\omega_0 t)\,\mathrm{d}t \neq 0$$

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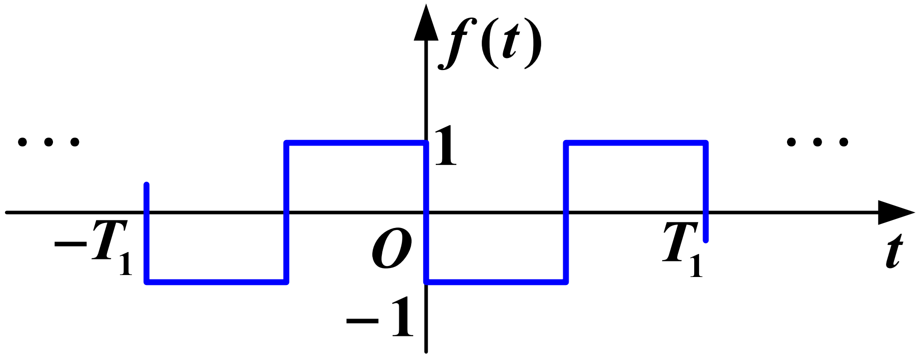
(2) Odd Function

The waveform is antisymmetric about the vertical axis $f(t) = -f(-t)$



$$a_0 = 0 \qquad a_k = 0 \qquad b_k \neq 0$$

$$F_k = \frac{1}{2}\left(a_k - \mathrm{j}b_k\right) = -\frac{1}{2}\mathrm{j}b_k$$

F_k is an imaginary function

Time-domain odd function → frequency-domain imaginary function

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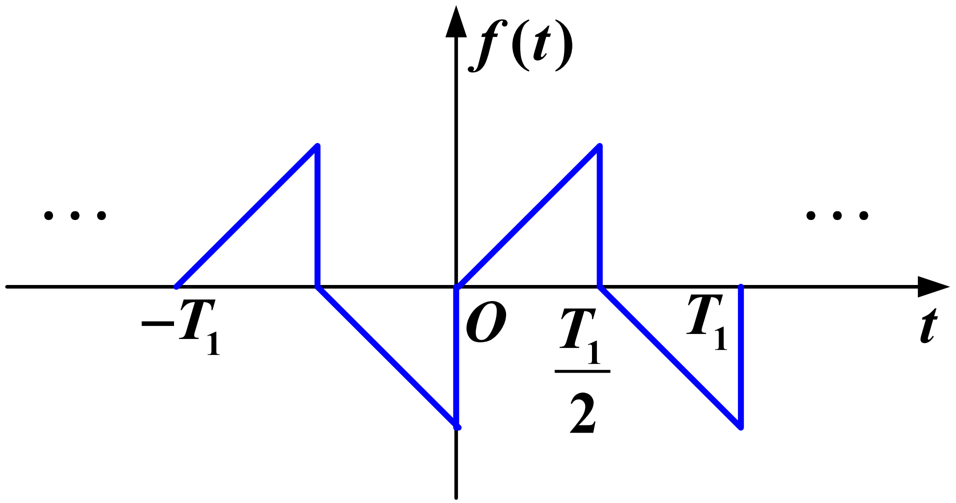
(3) Odd-Harmonic Function

A function is odd-harmonic (or possesses half-wave symmetry) if shifting it by half a period and inverting its amplitude leaves the waveform unchanged:

$$f(t) = -f\left(t \pm \frac{T_1}{2}\right)$$



$$a_0 = 0$$

$$k = 2, 4, 6 \cdots \qquad a_k = b_k = 0$$

Without even harmonic terms

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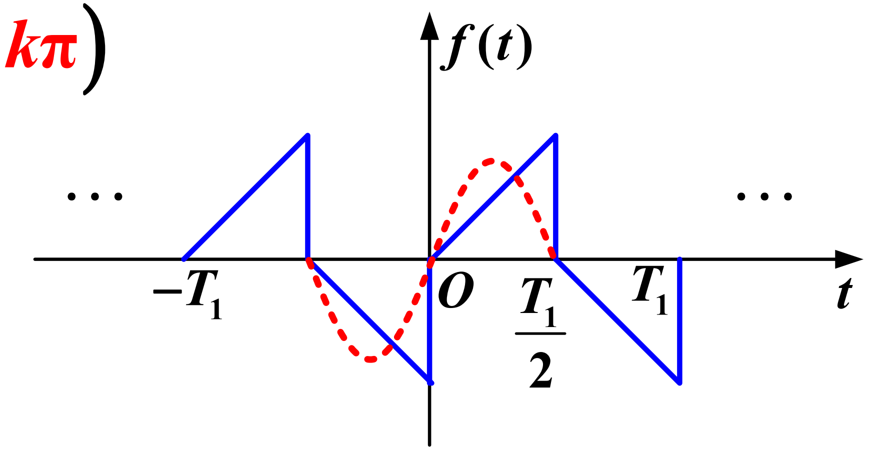
(3) Odd-Harmonic Function

Taking the sine terms as an example, the cosine-coefficient calculation is similar

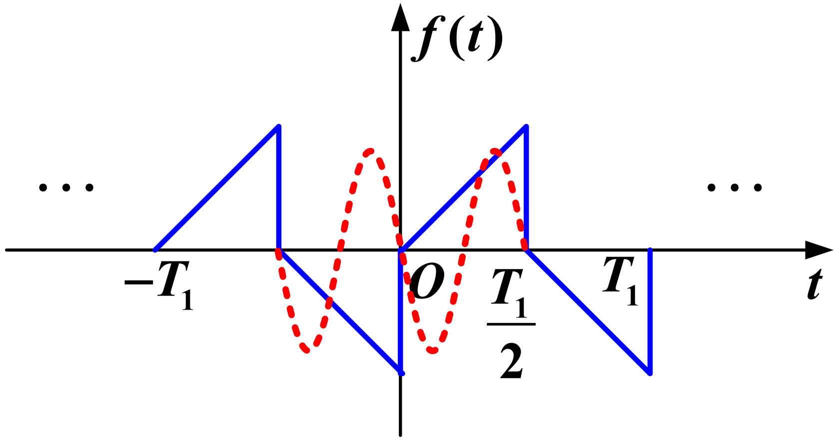
$$\sin\left[k\omega_0\left(t\pm\frac{T}{2}\right)\right]=\sin\left(k\omega_0 t\pm k\pi\right)$$

$$=\begin{cases}-\sin\left(k\omega_0 t\right) & k \text{ odd}\\ \sin\left(k\omega_0 t\right) & k \text{ even}\end{cases}$$



$$f\left(t\pm\frac{T}{2}\right)\sin\left[k\omega_0\left(t\pm\frac{T}{2}\right)\right]$$

$$=\begin{cases}f\left(t\right)\sin\left(k\omega_0 t\right) & k \text{ odd}\\ -f\left(t\right)\sin\left(k\omega_0 t\right) & k \text{ even}\end{cases}$$



Without even harmonic terms

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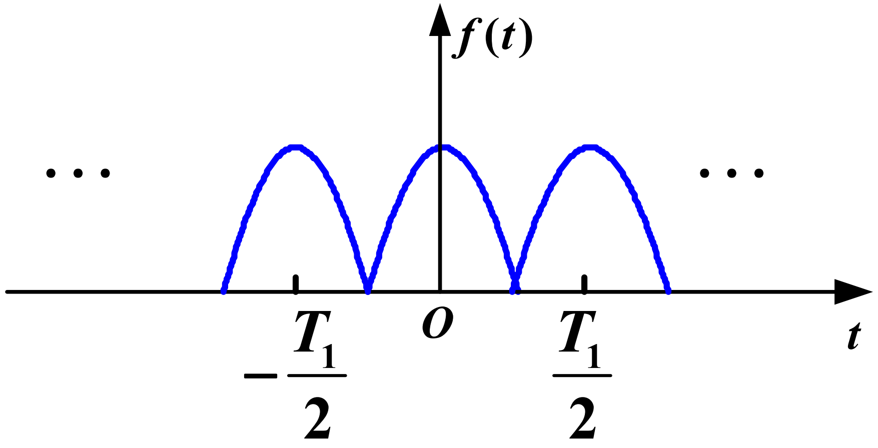
(4) Even-Harmonic Function

Effective Period is Half the Labeled Period

A function is even-harmonic if shifting it by half its labeled period $\pm\frac{T_1}{2}$ results in an identical waveform

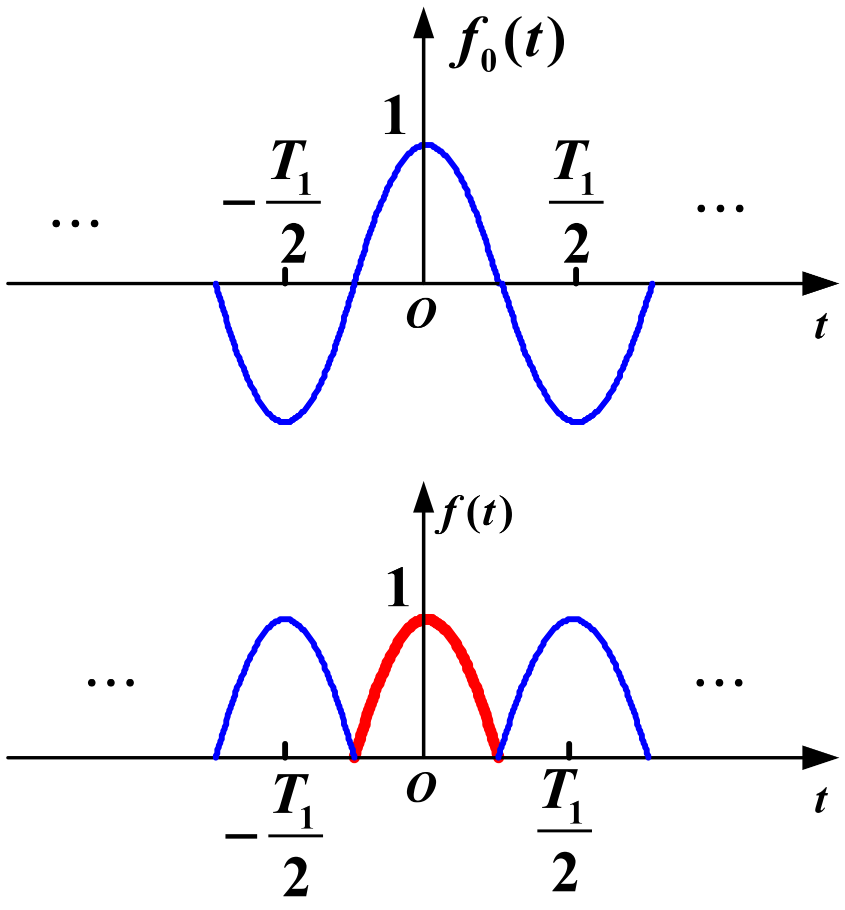
$$f(t) = f\left(t \pm \frac{T_1}{2}\right) \qquad \omega_1 = \frac{2\pi}{T_1}$$



$$\text{When } k = 1, 3, 5 \cdots \qquad a_k = b_k = 0$$

(4) Even-Harmonic Function

Effective Period is Half the Labeled Period



- T_1 — **Labeled period**

$\omega_1 = \frac{2\pi}{T_1}$ — **Fundamental angular frequency**

- $\frac{T_1}{2}$ — **Actual period**

The actual frequency components included are integer multiples of

$$\frac{2\pi}{\frac{T_1}{2}} = 2\omega_1$$

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4. Parseval’ s Theorem

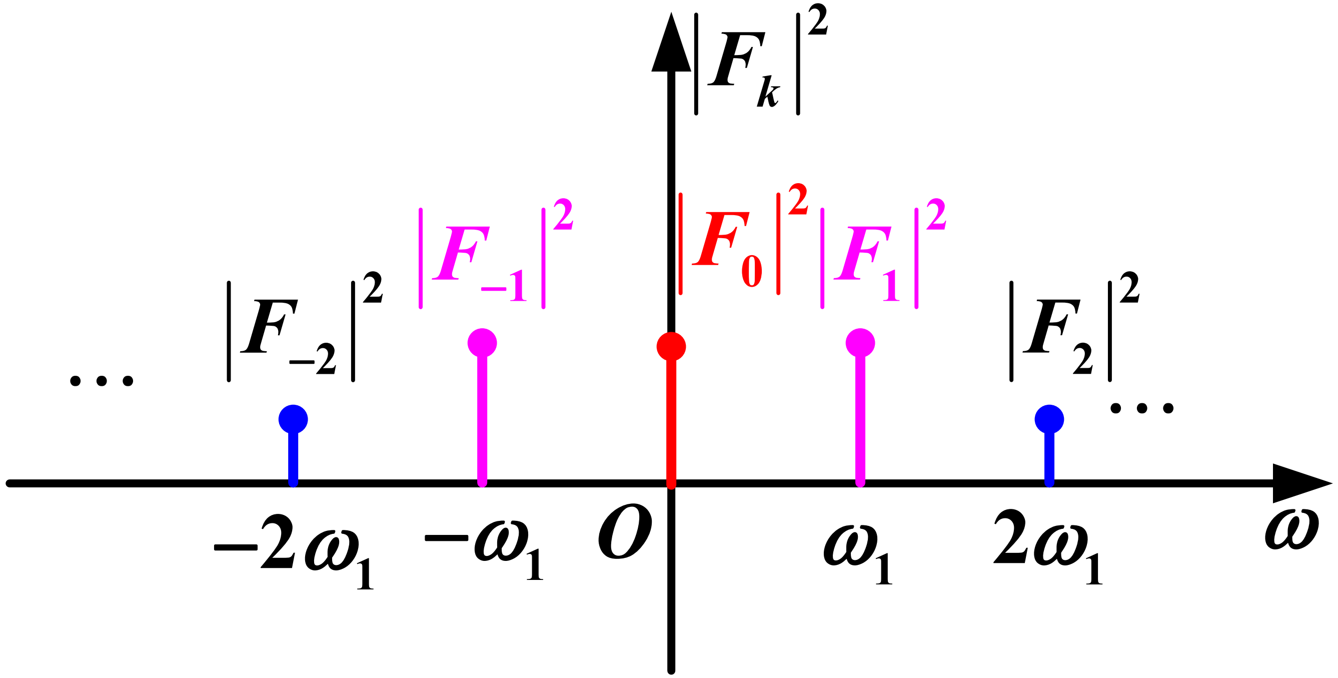
$$P = \frac{1}{T_1}\int_0^{T_1} \left| f(t) \right|^2 \mathrm{d}t = \begin{cases} a_0^2 + \frac{1}{2}\sum_{k=1}^{\infty}\left(a_k^2 + b_k^2\right) & \text{Trigonometric Form} \\ c_0^2 + \frac{1}{2}\sum_{k=1}^{\infty} c_k^2 & \text{Harmonic Form} \\ \sum_{k=-\infty}^{\infty} \left| F\left(k\omega_0\right) \right|^2 & \text{Exponential Form} \end{cases}$$

This is a specific manifestation of Parseval's theorem in the context of Fourier series

It indicates that: the average power of a periodic signal equals the sum of the squares of the effective values of the DC component, fundamental wave, and each harmonic component; in other words, energy is conserved between the time domain and the frequency domain

4. Parseval's Theorem

$|F_k|^2 \sim \omega$: **the distribution of the average power of each harmonic component with respect to frequency is called the power spectral coefficients**



5. Finite Fourier Series and Minimum Mean Square Error

$$f(t) = a_0 + \sum_{k=1}^{\infty}\left[a_k \cos(k\omega_0 t) + b_k \sin(k\omega_0 t)\right]$$

Approximate $f(t)$ by taking the first N harmonic terms

$$S_N(t) = a_0 + \sum_{k=1}^{N}\left[a_k \cos(k\omega_0 t) + b_k \sin(k\omega_0 t)\right]$$

Error function

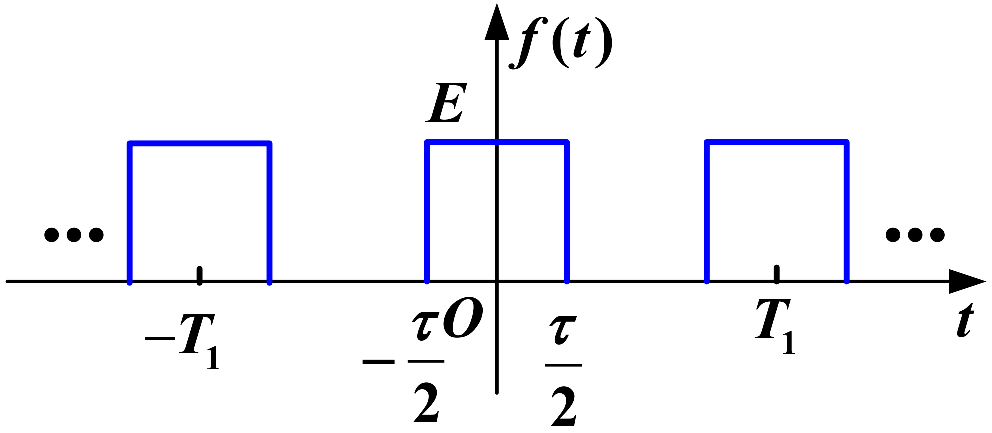
$$\varepsilon_N(t) = f(t) - S_N(t)$$

Mean Square Error

$$E_N = \overline{\varepsilon_N{}^2(t)} = P - \left[a_0{}^2 + \frac{1}{2}\sum_{k=1}^{N}\left(a_k{}^2 + b_k{}^2\right)\right]$$

6. Spectral Structure of Periodic Rectangular Pulse Trains

(1) Spectral Coefficients in Trigonometric Form



Pulse width τ

Period T_1

Amplitude E

$$a_0 = E\frac{\tau}{T_1} \quad \left(\frac{\tau}{T_1}:\text{Duty Cycle}\right)$$

$f(t)$ **is an even function** $\rightarrow b_k = 0 \quad F_k = \frac{1}{2}(a_k - \mathrm{j}b_k) = \frac{1}{2}a_k$

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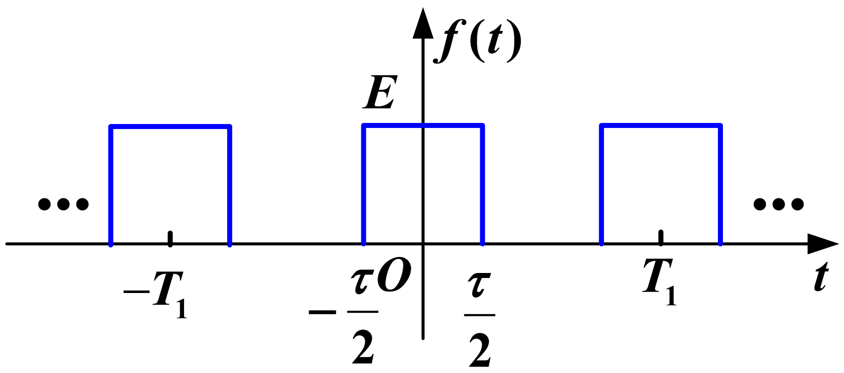
(2) Spectral Coefficients in Exponential Form

$$F_k = \frac{1}{T_1}\int_{-\frac{T_1}{2}}^{\frac{T_1}{2}} f(t)\mathrm{e}^{-\mathrm{j}k\omega_1 t}\,\mathrm{d}t$$

$$= \frac{1}{T_1}\int_{-\frac{\tau}{2}}^{\frac{\tau}{2}} E\mathrm{e}^{-\mathrm{j}k\omega_1 t}\,\mathrm{d}t$$

$$= \frac{E}{T_1}\frac{1}{-\mathrm{j}k\omega_1}\mathrm{e}^{-\mathrm{j}k\omega_1 t}\Big|_{-\frac{\tau}{2}}^{\frac{\tau}{2}}$$

$$= \frac{-E}{\mathrm{j}k\omega_1 T_1}\left(\mathrm{e}^{-\mathrm{j}k\omega_1\frac{\tau}{2}} - \mathrm{e}^{\mathrm{j}k\omega_1\frac{\tau}{2}}\right)$$



$$\omega_1 = \frac{2\pi}{T_1}$$

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(2) Spectral Coefficients in Exponential Form

$$F_k = \frac{-E}{\mathrm{j}k\omega_1 T_1}\left(\mathrm{e}^{-\mathrm{j}k\omega_1\frac{\tau}{2}} - \mathrm{e}^{\mathrm{j}k\omega_1\frac{\tau}{2}}\right)$$

$$= \frac{2E}{k\omega_1 T_1}\sin\left(k\omega_1\frac{\tau}{2}\right)$$

$$= \frac{E\tau}{T_1}\frac{\sin\left(k\omega_1\frac{\tau}{2}\right)}{k\omega_1\frac{\tau}{2}}$$

$$= \frac{E\tau}{T_1}\mathrm{Sa}\left(k\omega_1\frac{\tau}{2}\right)$$

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(3) Spectrum and Its Characteristics $(T_1 = 5\tau)$

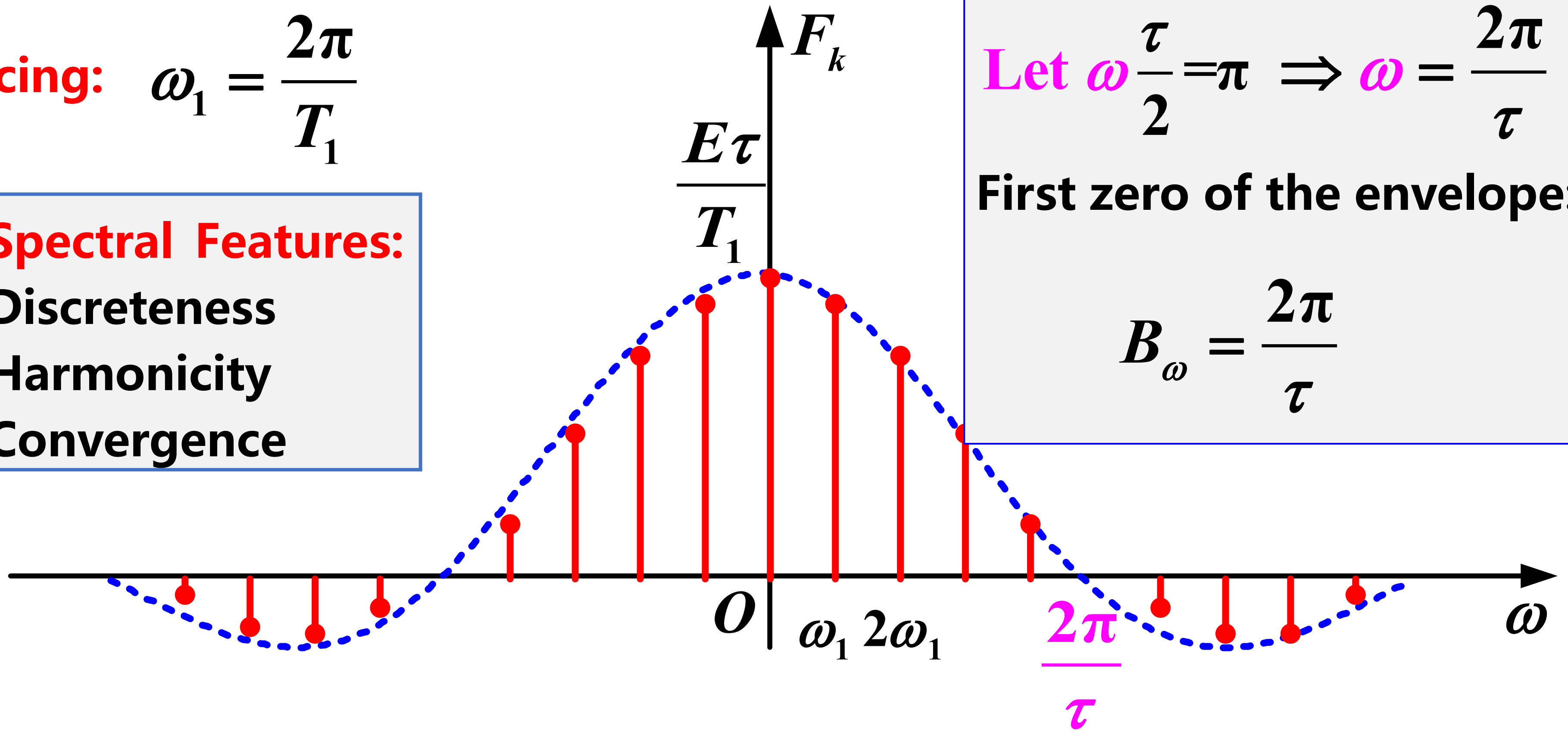
$$F_k = \frac{E\tau}{T_1}\mathrm{Sa}\left(k\omega_1\frac{\tau}{2}\right)$$

Spectral Coefficients: $\dfrac{E\tau}{T_1}\mathrm{Sa}\left(\omega\dfrac{\tau}{2}\right)$

Line spacing: $\omega_1 = \dfrac{2\pi}{T_1}$

Spectral Features:

- Discreteness
- Harmonicity
- Convergence

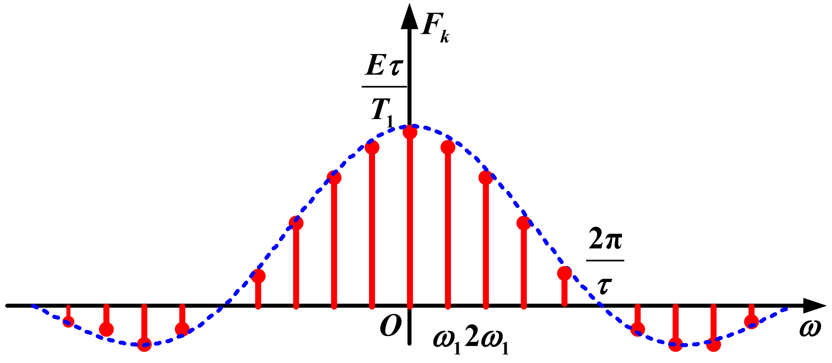


Let $\omega\dfrac{\tau}{2} = \pi \Rightarrow \omega = \dfrac{2\pi}{\tau}$

First zero of the envelope:

$$B_\omega = \frac{2\pi}{\tau}$$

(3) Spectrum and Its Characteristics

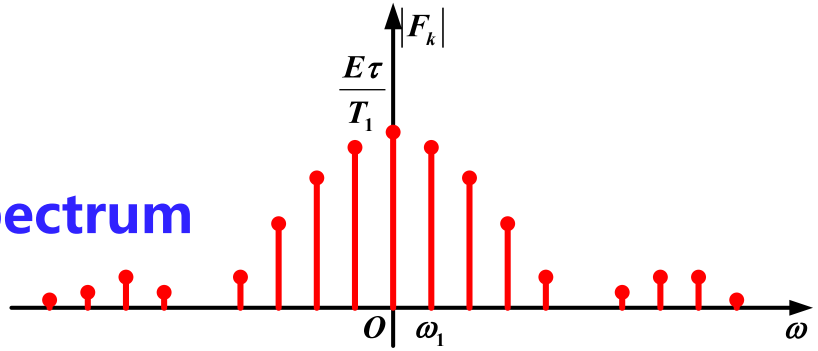


- $F_k > 0$:

$$F_k = |F_k| \cdot 1 = |F_k| \cdot e^{j0}$$

Phase is 0

Amplitude spectrum

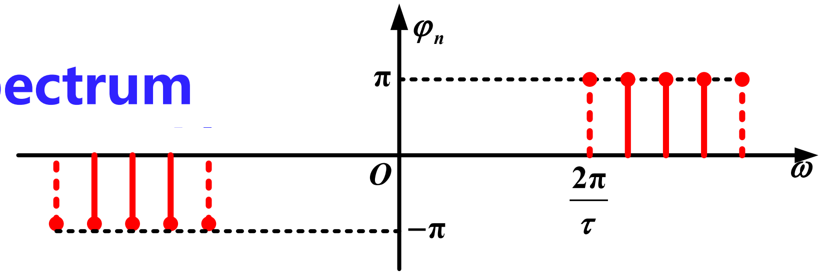


- $F_k < 0$:

$$F_k = |F_k| \cdot (-1) = |F_k| e^{j(\pm\pi)}$$

Phase is $\pm\pi$

Phase spectrum



(3) Spectrum and Its Characteristics

$$F_k = E\frac{\tau}{T_1}\mathrm{Sa}\left(k\omega_1\frac{\tau}{2}\right)$$

- The spectral coefficients F_k are directly proportional to E
- Pulse width τ: The first zero of the envelope occurs at: $B_\omega = \frac{2\pi}{\tau}$
  This bandwidth B_ω is inversely proportional to τ
  Time-domain compression → Frequency-domain expansion
- Period T_1 : The spectral line spacing is determined by: $\omega_1 = \frac{2\pi}{T_1}$
  As T_1 increases, the spectral line spacing decreases

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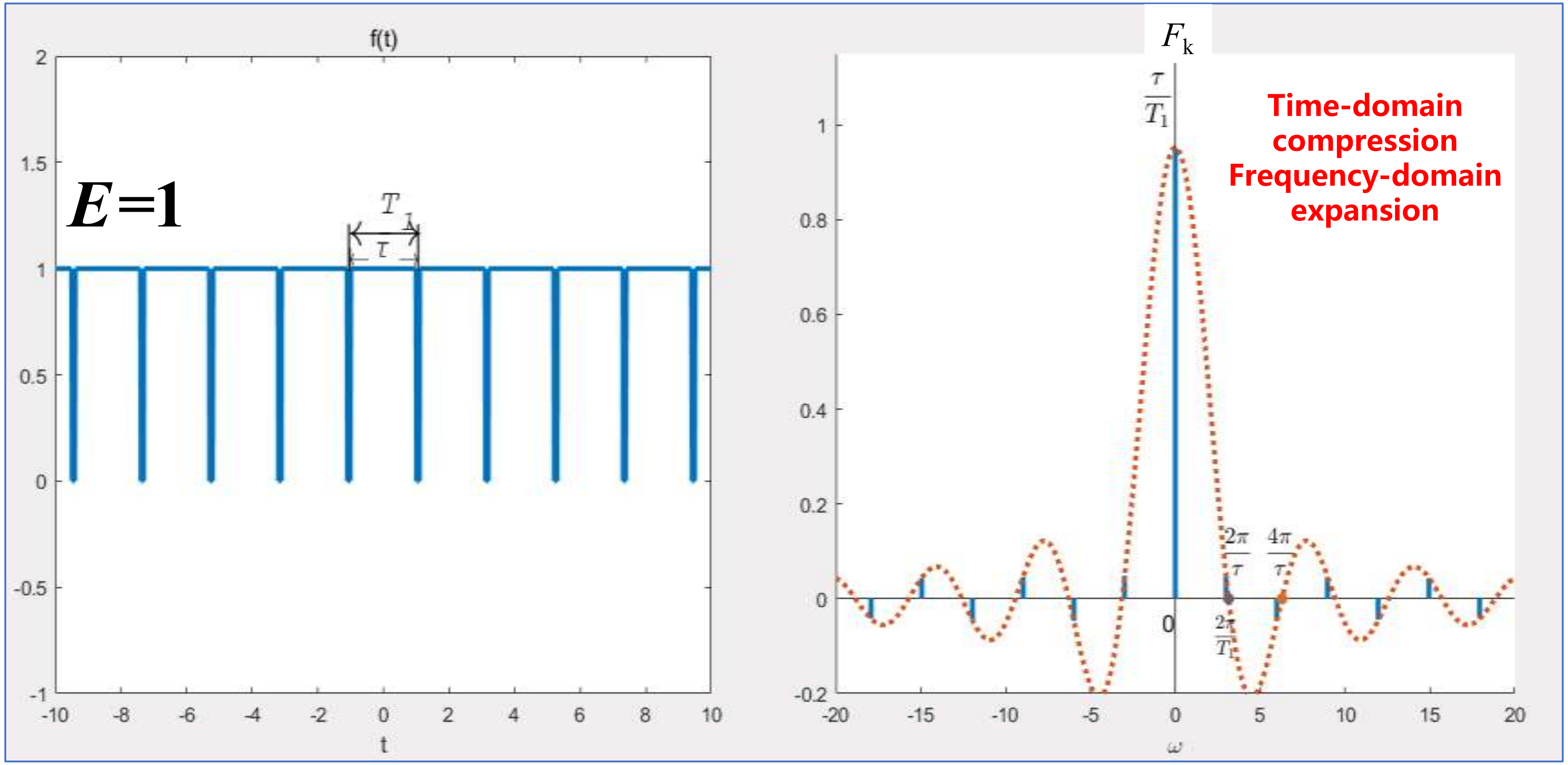
(3) Spectrum and Its Characteristics

$$F_k = E\frac{\tau}{T_1}\mathrm{Sa}\left(k\omega_1\frac{\tau}{2}\right)$$

T_1 unchanged, τ decreases $\Rightarrow$ $\omega_1=\frac{2\pi}{T_1}$ unchanged, $B_\omega=\frac{2\pi}{\tau}$ increases



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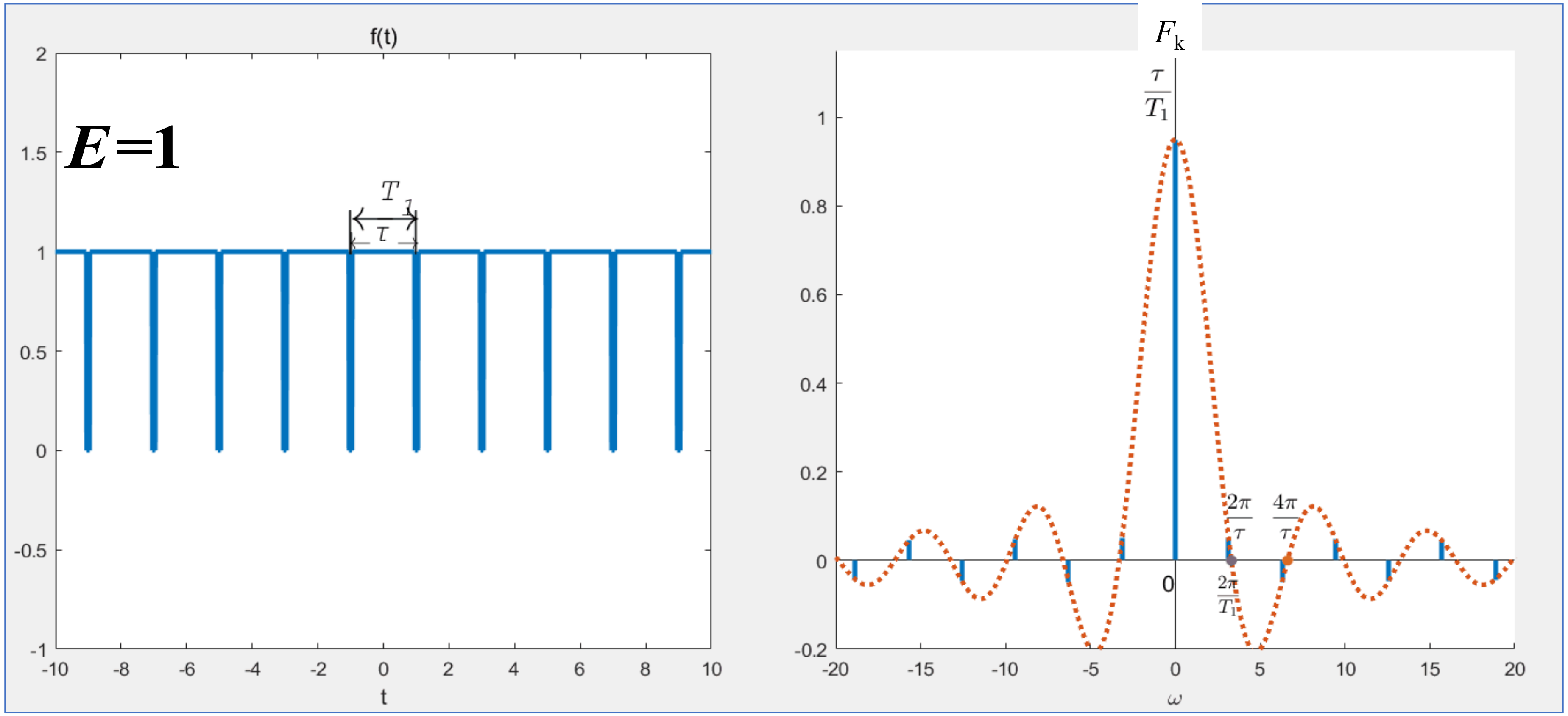
(3) Spectrum and Its Characteristics

$$F_k = E\frac{\tau}{T_1}\mathrm{Sa}\left(k\omega_1\frac{\tau}{2}\right)$$ τ unchanged, T_1 increases $\Rightarrow$ $\omega_1 = \frac{2\pi}{T_1}$ decreases, $B_\omega = \frac{2\pi}{\tau}$ unchanged



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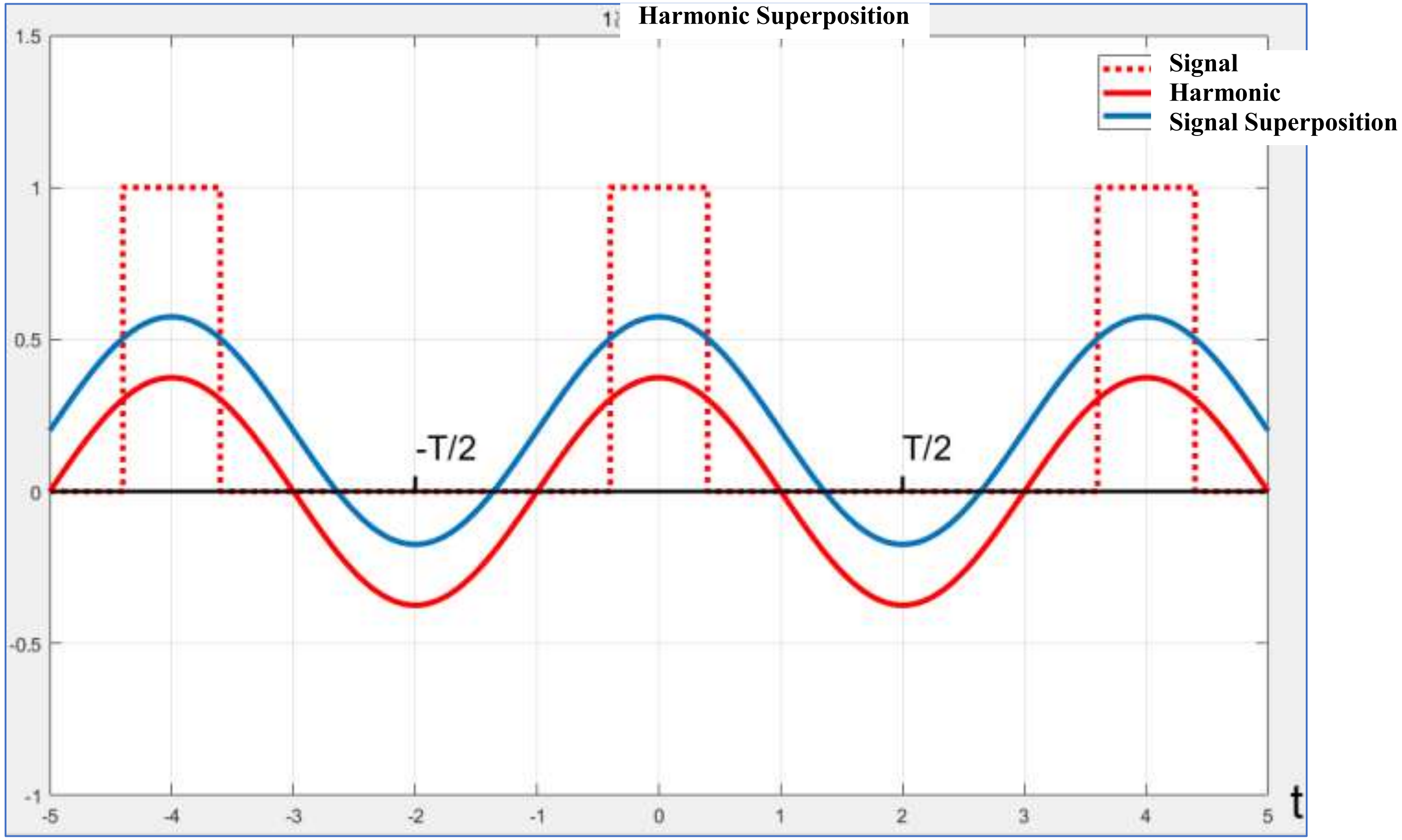
(4) Fourier Series Synthesis

$$f(t)=\sum_{k=-\infty}^{\infty} E\frac{\tau}{T_1}\mathrm{Sa}\left(k\omega_1\frac{\tau}{2}\right)\mathrm{e}^{\mathrm{j}k\omega_1 t} \quad \left(\omega_1=\frac{2\pi}{T_1}\right)$$

$$=E\frac{\tau}{T_1}+2E\frac{\tau}{T_1}\sum_{k=1}^{\infty}\mathrm{Sa}\left(k\omega_1\frac{\tau}{2}\right)\cos\left(k\omega_1 t\right)$$

(4) Fourier Series Synthesis

duty cycle of 20%



$$f(t)=E\frac{\tau}{T_1}+2E\frac{\tau}{T_1}\sum_{k=1}^{\infty}\mathrm{Sa}\left(k\omega_1\frac{\tau}{2}\right)\cos\left(k\omega_1 t\right)$$

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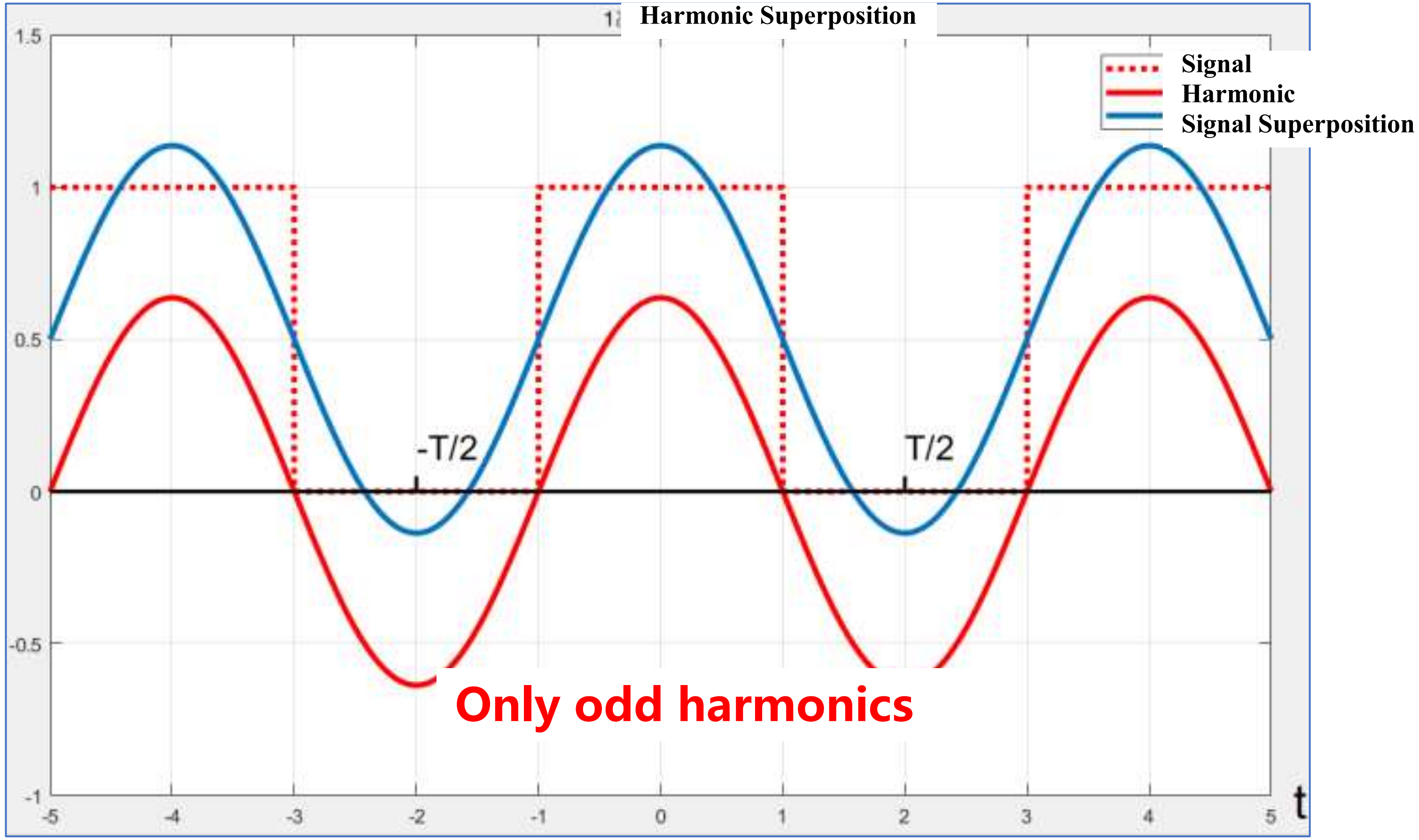
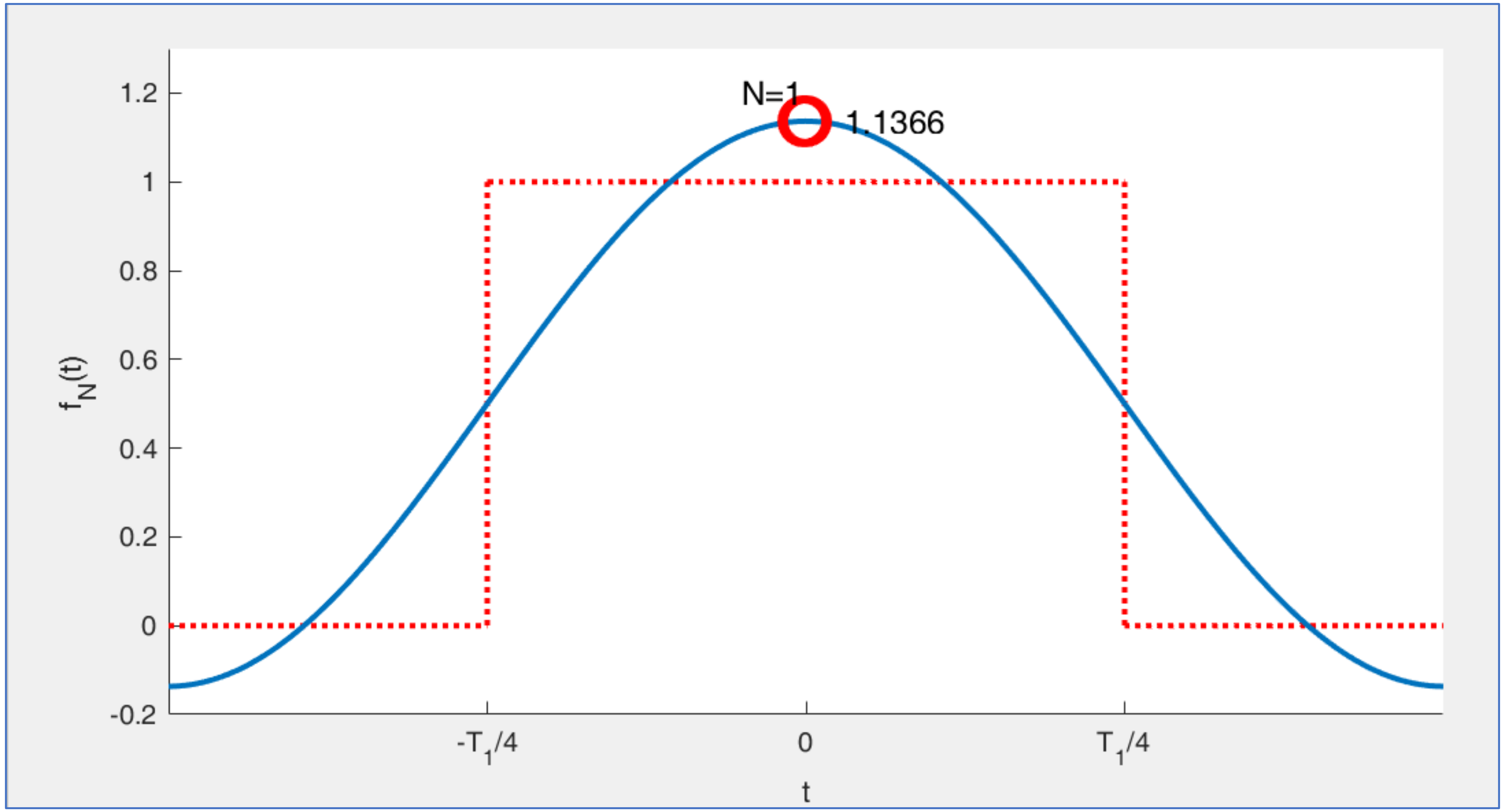
(4) Fourier Series Synthesis

duty cycle of 50%



$$f(t)=\frac{E}{2}+\frac{2E}{\pi}\left[\cos(\omega_1 t)-\frac{1}{3}\cos(3\omega_1 t)+\frac{1}{5}\cos(5\omega_1 t)+\cdots\right]$$

7. Gibbs Phenomenon



The more terms are included, the smaller the mean square error becomes, while the peak of the fluctuations approaches a constant value - approximately 9% of the total jump discontinuity. This is known as the Gibbs phenomenon

7. Gibbs Phenomenon

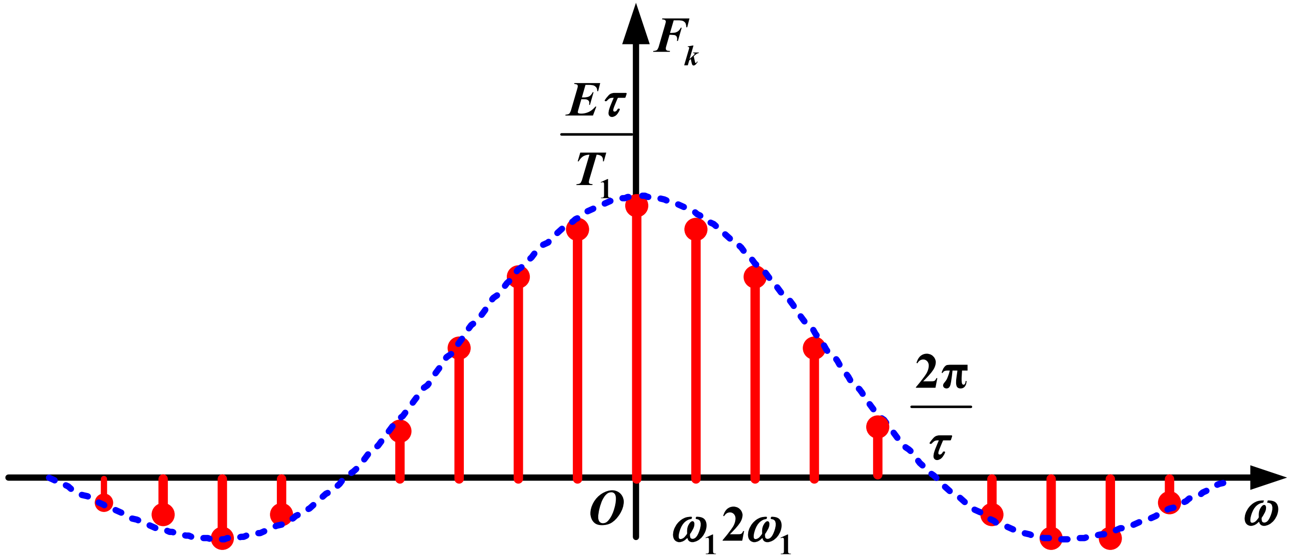
Fourier series decomposition follows the principle of minimizing mean square error in signal approximation. However, for signals with discontinuity points, the series fails to converge precisely at these discontinuities

In 1898, American physicist Albert Michelson observed this phenomenon while testing square wave signals using his self-designed harmonic analyzer. When attempting to synthesize signals, he noted persistent oscillations near discontinuities despite increasing the number of harmonic terms

The renowned mathematical physicist Josiah Gibbs investigated these findings and published his formal analysis in 1899

To understand its origin, we examine the ideal low-pass filter's behavior

8. Bandwidth



Under specified distortion conditions, a signal can be represented by components within a certain frequency range. This frequency range is called the bandwidth

$B_\omega = \frac{2\pi}{\tau}$ **:The bandwidth is defined by the first zero-crossing point of the signal**

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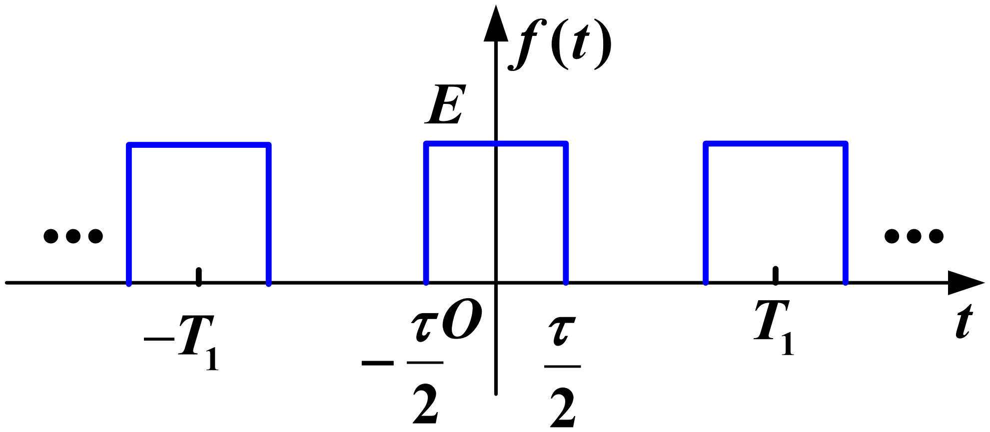
8. Bandwidth



Example: $\tau = \frac{1}{20}\,\mathrm{s}, T_1 = \frac{1}{4}\,\mathrm{s}$ the power of the signal is

$$P = \frac{1}{T_1}\int_{-\frac{T_1}{2}}^{\frac{T_1}{2}} f^2(t)\mathrm{d}t$$

$$= \frac{1}{T_1}\int_{-\frac{\tau}{2}}^{\frac{\tau}{2}} E^2\mathrm{d}t = 0.2E^2$$

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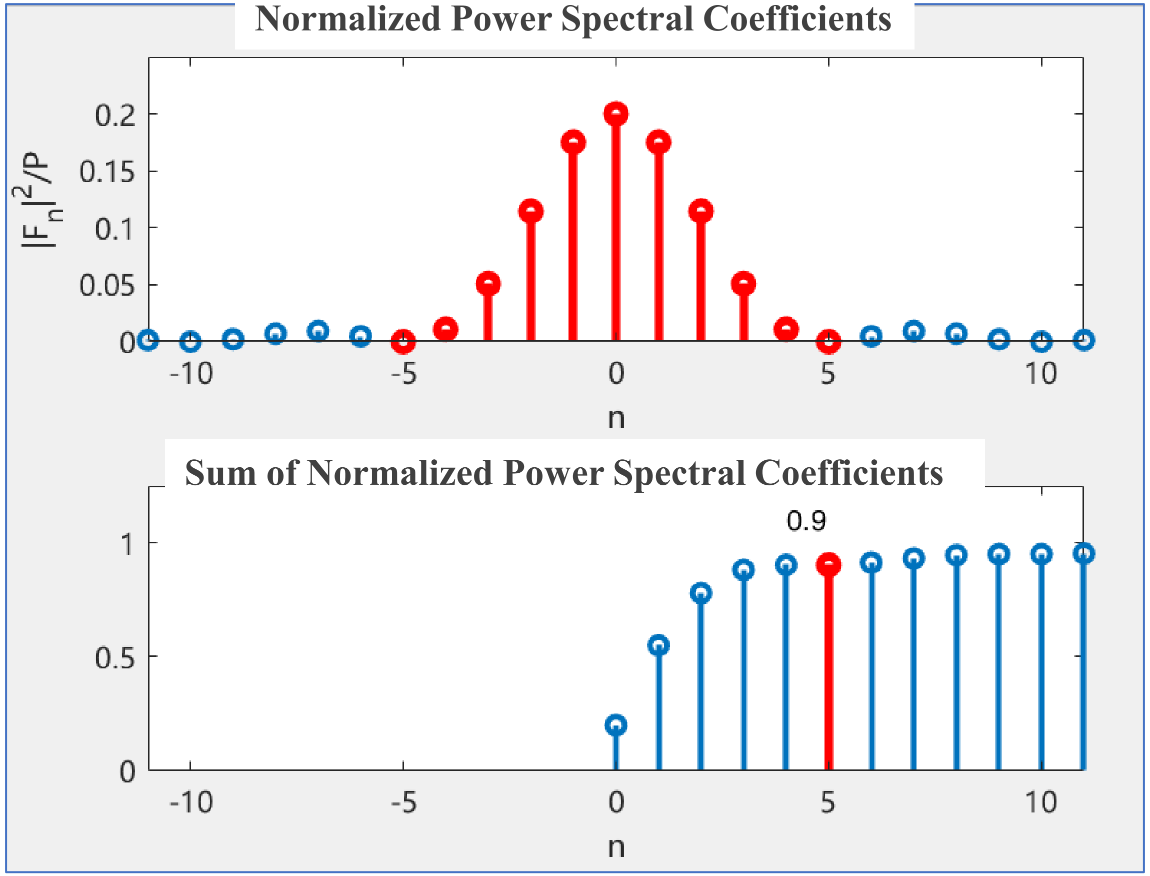
8. Bandwidth

The power of the signal within the main lobe of the spectrum is $P_5 = \sum_{n=-5}^{5} |F_k|^2 \approx 0.18E^2$

$$\frac{P_5}{P} \approx 90\%$$



8. Bandwidth

System's bandwidth > Signal's bandwidth, to avoid distortion

Speech Signal	300~3400Hz
Music Signal	50~15,000Hz
Loudspeaker	15~20,000Hz

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Summary

(1) The Fourier series of a periodic signal $f(t)$ has two forms

Trigonometric Form

$$f(t)=a_0+\sum_{k=1}^{\infty}\left[a_k\cos\left(k\omega_1 t\right)+b_k\sin\left(k\omega_1 t\right)\right]$$

$$=c_0+\sum_{k=1}^{\infty}c_k\cos\left(k\omega_1 t+\varphi_k\right)$$

Exponential Form

$$f(t)=\sum_{k=-\infty}^{\infty}F_k\mathrm{e}^{\mathrm{j}k\omega_1 t}$$

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Summary

(2) Relationship Between the Two Forms of Fourier Series

- The trigonometric form: $c_k \sim \omega, \quad \varphi_k \sim \omega$ One-sided spectrum
- The exponential form: $\left|F_k\right| \sim \omega, \quad \varphi_k \sim \omega$ Double-sided spectrum

Relationship $\left|F_k\right| = \frac{1}{2} c_k \left(k \neq 0\right)$ $F_0 = c_0 = a_0$

- In the exponential form, the amplitude spectrum is an even function

$$\left|F_k\right| = \left|F_{-k}\right|$$

- The phase spectrum is an odd function

$$\varphi_k = -\ \varphi_{-k}$$

Summary

(3) Three Properties

Discreteness: Spectral lines

Harmonicity: Frequency components occur only at $k\omega_1$

Convergence: $|k|\uparrow, \quad |F_k|\downarrow$

Note: These properties are fundamental to analyzing periodic signals but may not hold for non-periodic or singular signals (e.g., ideal impulse trains)

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Summary

(4) Introduction of Negative Frequencies

In two-sided spectra, negative frequencies ($n\omega_1$) are mathematical constructs without direct physical interpretation

Why Include Negative Frequencies?

$f(t)$ is a real function, decomposed into complex exponentials, and it must have a conjugate pair of $e^{jk\omega_1}$ and $e^{-jk\omega_1}$ to preserve the properties of being a real function

$$c_k \cos\left(k\omega_1 t + \varphi_k\right) = \frac{1}{2} c_k \left[\mathrm{e}^{\mathrm{j}\left(k\omega_1 t + \varphi_k\right)} + \mathrm{e}^{-\mathrm{j}\left(k\omega_1 t + \varphi_k\right)} \right]$$

Summary

(5) Spectrum of Typical Signals

- Spectrum of Periodic Rectangular Pulse Trains
- Gibbs Phenomenon
- Bandwidth